

# Afferent Health

Clinical data and ML/AI decision support for high-stakes care and federal health missions.

## CAPABILITY STATEMENT

Afferent Health LLC  
New York, NY

### WHO WE ARE

**Afferent Health LLC** is an independent scientific and clinical R&D intelligence firm. We help health systems, federal partners, and technology teams turn complex clinical data into deployable decision-support capabilities by combining clinician-led problem definition, advanced ML/AI development, and specialized execution expertise.

### CORE CAPABILITIES

- **Clinical data and multimodal physiology.** Cohort design and integration across waveforms, monitor numerics, EHR data, and laboratory results.
- **ML/AI decision support.** Prototype development, calibration, subgroup testing, explainability, safety analysis, and test/evaluation/verification/validation support.
- **Perioperative, ICU, and acute-care intelligence.** Risk stratification and monitoring for deterioration, delirium, respiratory events, surgical outcomes, and care navigation.
- **Research-to-product translation.** Evidence synthesis, economic-impact modeling, product scoping, study design, and proof-of-concept planning.
- **Federal health solution shaping.** Clinical requirements translation, RFI/RFP content, bid/no-bid analysis, work-package design, and teaming support.
- **Specialized clinical and scientific expertise.** Clinicians and AI/ML scientists embedded where mission delivery requires domain depth.

### DIFFERENTIATORS

- **Clinician-scientist led.** Clinical validity, workflow fit, and measurable impact are designed in from the start.
- **One bridge from data to procurement.** Afferent connects clinical need, research design, model validation, economic rationale, and federal capture.
- **Partner-enabled delivery.** Afferent supplies clinical and scientific intelligence while its partner ecosystem extends engineering, cloud, integration, and federal program delivery.

### PARTNER ECOSYSTEM



### COMPANY DATA

**LEGAL ENTITY**  
Afferent Health LLC  
New York, NY

**PRIMARY NAICS**  
541690  
Other Scientific and Technical Consulting Services

**ADDITIONAL NAICS**  
541715 — R&D in Physical, Engineering, and Life Sciences  
541512 — Computer Systems Design Services

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*Afferent leads clinical and scientific intelligence; partners extend engineering, infrastructure, integration, and federal delivery.*

### KEY PERSONNEL

- **D. K. Adrian Williams, MSc — Founder & Managing Principal.** Physician-scientist in training and clinical modeling lead working in multimodal physiology, pulse oximetry, capnography, and EEG.
- **Joseph R. Scarpa, MD-PhD — Medical Consultant.** Anesthesiologist and NIA-funded investigator in perioperative machine learning, biological aging, and surgical outcomes.
- **Ruben Coen-Cagli, PhD — Scientific/AI Consultant.** Machine-learning methods investigator focused on high-dimensional signals, computational neuroscience, and rigorous model development.

### CURRENT WORK AND EVIDENCE

- **Multimodal perioperative and ICU modeling.** Clinical-risk concepts built around high-resolution physiologic and EHR data.
- **Physiologic-signal methods.** Active work spanning pulse oximetry, capnography, EEG, and accountable clinical AI.
- **Product scoping.** Surgical biological age, delirium, transplant decision support, and clinical AI assurance/TEVV concepts.
- **Federal capture intelligence.** Structured opportunity monitoring, vehicle-fit analysis, and clinical work-package design for health missions.

### ENGAGEMENT MODEL

1. **Define.** Identify the documented clinical, operational, and economic problem.
2. **Demonstrate.** Use available data and prior work to establish technical and financial plausibility.
3. **Deliver.** Package the clinical workstream for a pilot, task order, grant, or partner-led federal response.